



SmolVLM: Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University

Competitive vision-language models at sub-1GB RAM



Large VLMs Are Powerful but Impractical for Edge Deployment

Current State

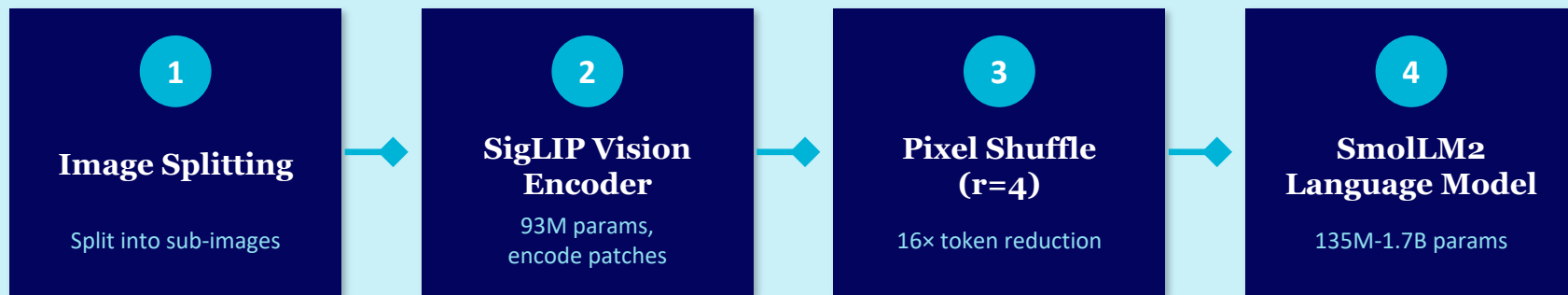
- State-of-the-art VLMs: 7B-80B parameters
- GPU RAM: 27+ GB required
- Cloud-dependent inference

Edge Reality

- Mobile devices: 4-8 GB total RAM
- Latency-sensitive applications
- Privacy: on-device processing needed

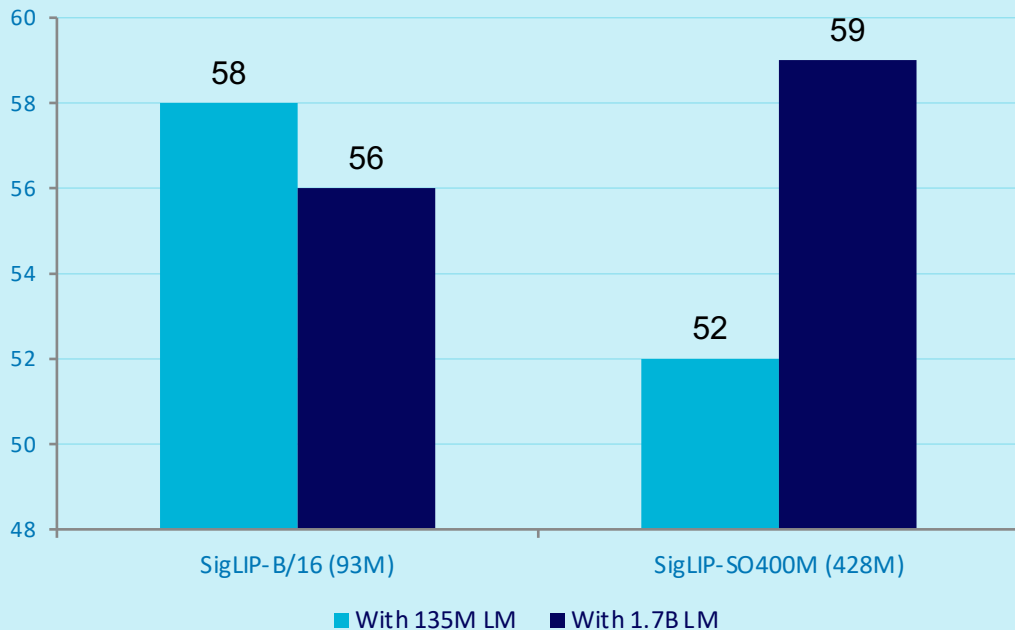
SmolVLM-256M runs at < 1 GB RAM — outperforming the 300× larger Llafacs-80B

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone



Images are split, encoded, compressed via pixel shuffle, and projected into SmolLM2's input space

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs



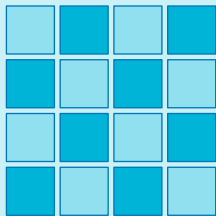
Finding 1

The 93M SigLIP-B/16 encoder outperforms the 428M SigLIP-SO400M for the smallest language models.

Only at 1.7B LM scale does the larger encoder justify its 10% parameter overhead.

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

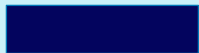
Before Pixel Shuffle



H × W spatial tokens



After Pixel Shuffle (r=4)



H/4 × W/4 tokens, r² channels

16×
fewer tokens

r = 2 (standard)

4× token reduction

Best for larger models (2B+)

r = 4 (SmolVLM)

16× token reduction

Best for small models (<1B)

Aggressive compression eases attention overhead and improves long-context modeling

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

RoPE Base Adjustment

Base frequency increased from 10K → 273K to support longer sequences

10K → 273K

Fine-tuning Strategy

Mixed long/short-context data ensures robust performance across all sequence lengths

Mixed Data

Model-Specific Context

SmolVLM-2.2B → 16K tokens
SmolVLM-256M/500M → 8K tokens

Up to 16K

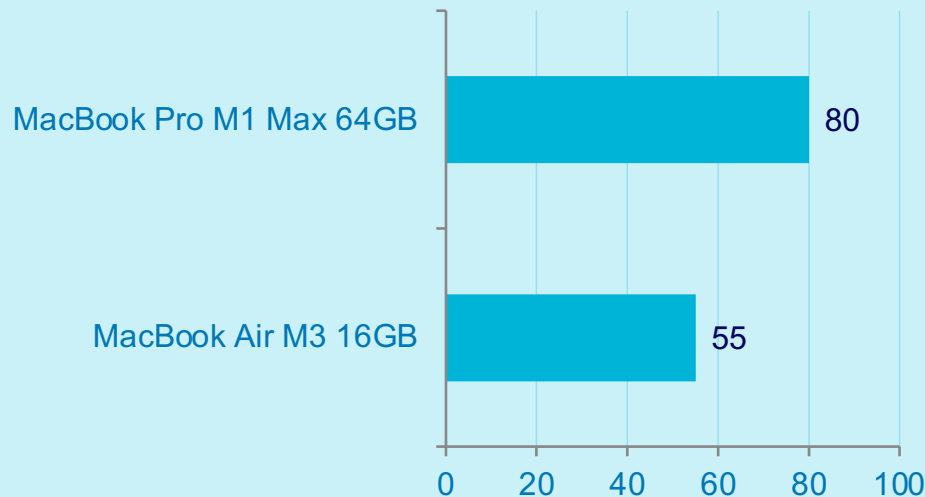
Performance improves significantly when extending context from 2K to 16K tokens

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Capability	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo)
	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo)
	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo)
	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo)
	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo)
	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo)
	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo)
	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2)
	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2)
Average	All	44.0%	51.0%	59.8%	—

RAM Usage: 256M → 0.8 GB | 500M → 1.2 GB | 2.2B → 4.9 GB | Molmo-A1B-7B → 27.7 GB

Real-Time Inference on Apple Silicon: Up to 80 Tokens/Second



HuggingSnap iOS App

Fully local inference, no cloud dependency

Batch Scaling

At batch 64: ~1000 tok/s on high-end devices

Linear Throughput

Throughput scales linearly with batch size for 256M

Key Takeaways: Design Principles for Efficient Multimodal Models

- 1 Balance encoder-LM compute: 93M SigLIP-B/16 outperforms 428M SigLIP-SO400M for smallest LMs
- 2 Use aggressive token compression: pixel shuffle $r=4$ yields 16 $\times$ reduction for sub-1B models
- 3 Extend context length: RoPE base 10K $\rightarrow$ 273K enables 16K-token context for higher resolution
- 4 Sub-1GB inference is achievable: SmolVLM-256M uses only 0.8 GB, surpassing 300 $\times$ larger LLaVA-80B
- 5 Fully open-source: all weights, data, code, and HuggingSnap mobile app released for reproducibility

Implications: These principles enable practical on-device multimodal AI deployment